

Data Migration Analysis Report — SAP S/4HANA

Optimization Strategy: LSMW × BAPI × Migration Cockpit

Date: 2026-07-30 **Client / Project:** ConcentoRDG **Scope:** Analysis of 7 data load objects currently running on LSMW / Migration Cockpit / T-code **Objective:** Reduce the total cutover load window (currently ~186h) through selective refactor to BAPIs with parallelism and package-level commits

1. Executive Summary

Indicator	Current	Projected	Reduction
Total load window	186 h	94 h – 131 h	30% – 50%
Objects on LSMW/IDoc	6	1 (optional)	–
Objects on parallel BAPI	0	5	–
Objects on Migration Cockpit	1	1 – 2	–
Manual T-code	1	1	–

Primary recommendation: prioritize refactor via BAPI + parallelism on **3 objects** that account for ~65% of the achievable savings:

1. Material Equipment Conversion (40h → 10–20h)
2. Open Production Order (60h → 36–48h)
3. Open PO Conversion (30h → 12–18h)

2. Full Object Inventory

#	Track	WRICEF	Object	Current method	Time (h)	Approx. total volume
1	SC	ZCON1091	Inventory (EWM)	T-code /SCWM/ISU	24	28,687
2	S2P	ZCON275	Open Purchase Orders	LSMW/IDoc	12	~838,000
3	S2P	ZCON033	Vendor Open Items	LSMW/IDoc	10	~465,000
4	R&M	ZCON186	Maintenance Work Order	Migration Cockpit	10	~2,850,000
5	PTS	PTS-20028-C	Open Production Order	LSMW multipart	60	~6,300,000
6	PTS	PTS-20041-C	Material Equipment Conversion (from MFC ECC)	LSMW/IDoc	40	~59,200,000

#	Track	WRICEF	Object	Current method	Time (h)	Approx. total volume
7	PTP	PTP-00035-C	Open PO Conversion	LSMW/IDoc	30	~4,000,000
TOTAL					186	~72.7 million

3. Detailed Technical Mapping

3.1 Inventory (EWM) — 24h

- **Target tables:** [/SCWM/ACQUA](#) (and related [/SCWM/QUAN](#), [/SCWM/ORDIM_*](#))
- **Equivalent BAPI:** *No classic BAPI available*
- **Cloud-ready alternatives:**
 - APIs [/SCWM/API_PHYSTOCK_CREATE](#), [/SCWM/ERP_STOCK_CREATE](#)
 - For MM-IM (non-EWM): [BAPI_GOODSMVT_CREATE](#) (movement 561)
- **Migration Cockpit — standard object: "Warehouse stock (EWM)"** (technical name [S4_EWM_STOCK](#) / [EWM_STOCK](#) depending on release)
- **Related EWM objects available in the Cockpit:**
 - Physical inventory document (EWM)
 - Warehouse product (EWM)
 - Storage bin (EWM)
 - Fixed bin assignment (EWM)
 - Handling Unit (EWM)
- **Recommendation:** keep [/SCWM/ISU](#) or migrate to Migration Cockpit. Gain from switching method is low (0–20%). Best leverage comes from:
 - Parallelism per warehouse / storage type
 - Increase work processes during the window
 - Reduce scope (current balance only, no movement history)

3.2 Open Purchase Orders — 12h

- **Target tables:** EKKO (6k), EKPO (18k), EKKN (4k), EKET (12k), EKPA (25k), STXH (27k), STXL (302k), ESLH (155k), ESLL (181k)
- **Recommended BAPI:** [BAPI_PO_CREATE1](#)
- **Table coverage:**

Table	Covered by BAPI?	How
EKKO	Yes	Parameter POHEADER
EKPO	Yes	Table POITEM
EKKN	Yes	Table POACCOUNT
EKET	Yes	Table POSCHEDULE
EKPA	Yes	Table POPARTNER
STXH / STXL	Yes	Tables POITEM_TEXT / POTEXTHEADER

Table	Covered by BAPI?	How
ESLH / ESLL	Yes	Tables POSERVICES + POSRVACCESSVALUES

- **Coverage: High**
- **Migration Cockpit — object:** "Purchase Order"

3.3 Vendor Open Items — 10h

- **Target tables:** BKPF (216k), BSEG (234k), WITH_ITEM (13k)
- **Recommended BAPI:** **BAPI_ACC_DOCUMENT_POST** or **BAPI_ACC_AP_DOCUMENT_POST**
- **Table coverage:**

Table	Covered by BAPI?	How
BKPF	Yes	Parameter DOCUMENTHEADER
BSEG	Yes	Tables ACCOUNTGL , ACCOUNTPAYABLE , CURRENCYAMOUNT
WITH_ITEM	Yes	Structures EXTENSION1 / EXTENSION2 with withholding fields

- **Coverage: High**
- **Migration Cockpit — object:** "Open items in AP / Vendor open items"

3.4 Maintenance Work Order — 10h (already in Cockpit)

- **Target tables:** AUFK (50k), AFVC (171k), RESB (36k), OBJK (105k), TEXT_SAP_AFVC (2.58M)
- **Equivalent BAPI:** **BAPI_ALM_ORDER_MAINTAIN** (modern, replaced **BAPI_ALM_ORDER_CREATE/_CHANGE**)
- **Table coverage:**

Table	Covered by BAPI?
AUFK	Yes (header)
AFVC	Yes (operations)
RESB	Yes (components)
OBJK	Yes (object list)
TEXT_SAP_AFVC	Yes (parameter LONGTEXTS)

- **Migration Cockpit — object:** "Maintenance order" (currently in use)
- **Recommendation:** keep Migration Cockpit. Gain from switching to BAPI is marginal (10–20%).

3.5 Open Production Order — 60h 🚩 largest bottleneck

- **Target tables:** AFKO (443k), AFPO (443k), AFVC (1M), AFVV (1M), AFRU (350k), AFFL (60k), RESB (1.74M), OBJK (105k), PEG_TASS (51k), SER05 (38k), TEXT_SAP_AUFK (89k)
- **Recommended BAPIs:**
 - **BAPI_PRODORD_CREATE** — creates header + operations + components
 - **BAPI_PRODORD_CHANGE** — complements fields not covered on creation
 - **BAPI_PRODORDCONF_CREATE_TT** / **BAPI_PRODORDCONF_CREATE_HDR** — confirmations (AFRU)
- **Table coverage:**

Table	Covered by BAPI?	Notes
AFKO, AFPO, AFVC, AFVV, AFFL, RESB	Yes	via BAPI_PRODORD_CREATE
AFRU	Yes	via confirmation BAPI (separate)
PEG_TASS (pegging)	No	Custom Z update or specific API
SER05 (serial numbers)	No	BAPI_SERNR_ADD_TO_DOCUMENT
TEXT_SAP_AUFG	Yes	via SAVE_TEXT in batch

- **Coverage: Partial**
- **Recommendation:** proof of concept before deciding. The current *Child Order LSMW* takes ~40h — if **BAPI_PRODORD_CREATE** covers the fields in use, the gain can be substantial; otherwise, keep LSMW.

3.6 Material Equipment Conversion (from MFC ECC) — 40h ⚠️ largest volume

- **Target tables:** EQUZ (38,457,154), EQUI (20,767,496)
- **Recommended BAPI:** **BAPI_EQUI_CREATE** + **BAPI_EQUI_CHANGE** (or **BAPI_EQMI_CREATE** to install at a Functional Location)
- **Table coverage:**

Table	Covered by BAPI?
EQUI	Yes (master data)
EQUZ	Yes (usage period / installation)

- **Coverage: High**
- **Migration Cockpit — object:** "Equipment"
- **Recommendation: top refactor candidate.** Massive volume (~59M) + mature BAPI + tables fully covered. Also consider scope reduction on EQUZ (current status only, no full history) — this may save more hours than any technical optimization.

3.7 Open PO Conversion — 30h

- **Target tables:** EKKO_PO (28k), EKPO_PO (49k), EKKN_PO (50k), EKET_PO (80k), RESB_PO (4k), ESLL_PO (78), STXL_PO (3M), **ZLPO_WBS_BRKDWN (125k)**, **ZSC_PO_TEXT_KEY (612k)**, **PRCD_ELEMENTS_PO (19k)**
- **Recommended BAPI:** **BAPI_PO_CREATE1**
- **Table coverage:**

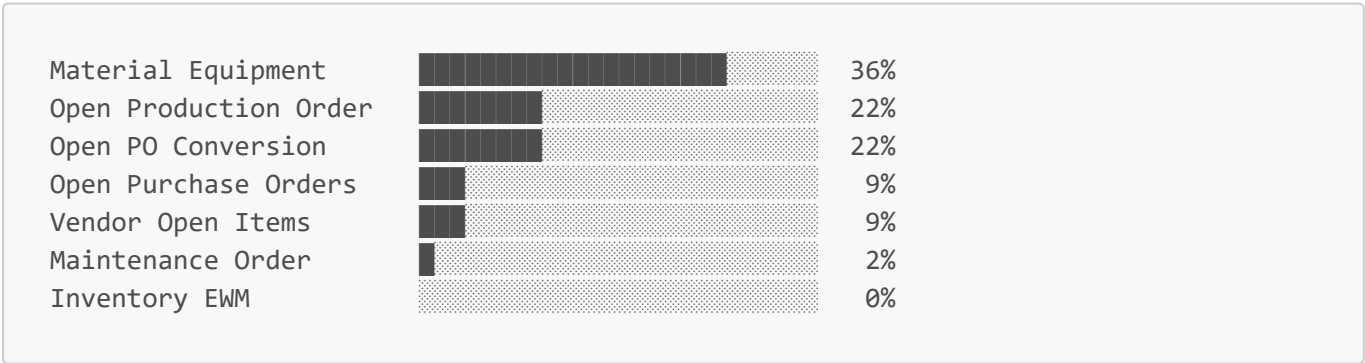
Table	Covered by BAPI?	Notes
EKKO_PO / EKPO_PO / EKKN_PO / EKET_PO / RESB_PO / ESLL_PO / STXL_PO	Yes	via standard BAPI_PO_CREATE1
ZLPO_WBS_BRKDWN	No	Custom Z table — post-BAPI update
ZSC_PO_TEXT_KEY	No	Custom Z table — post-BAPI update
PRCD_ELEMENTS_PO	Partial	If manual conditions → POCOND/POCONDX . If determined by pricing procedure → let SAP

Table	Covered by BAPI?	Notes
		recalculate
• Coverage: Medium • Recommendation: BAPI + post-commit Z routine for the Z tables. Estimate already accounts for this additional effort.		

4. ROI Analysis — Potential Gain

#	Object	Current (h)	Realistic gain	Projected (h)	Savings (h)	% Savings
6	Material Equipment	40	50–75%	10–20	20–30	60%
5	Open Production Order	60	20–40%	36–48	12–24	30%
7	Open PO Conversion	30	40–60%	12–18	12–18	50%
2	Open Purchase Orders	12	40–60%	5–7	5–7	50%
3	Vendor Open Items	10	50–70%	3–5	5–7	60%
4	Maintenance Work Order	10	10–20%	8–9	1–2	15%
1	Inventory EWM	24	0–20%	20–24	0–4	10%
TOTAL		186		94–131	55–92	30–50%

Savings distribution by object (conservative scenario ~55h)



5. Suggested Implementation Plan

Phase 1 — Quick wins (ROI-focused)

Sprint	Object	Deliverable	Expected gain
S1	Material Equipment	Z program with <code>BAPI_EQUI_CREATE</code> + package 500 + parallelism	20–30h

Sprint	Object	Deliverable	Expected gain
S1	Vendor Open Items	Z program with BAPI_ACC_DOCUMENT_POST + package 200 + parallelism	5–7h
S2	Open PO Conversion	Z program with BAPI_PO_CREATE1 + post-commit Z update	12–18h
S2	Open Purchase Orders	Reuse code from S2	5–7h

Phase 2 — Technical evaluation

Sprint	Object	Deliverable
S3	Open Production Order	POC with BAPI_PRODORD_CREATE to validate field coverage and real gain

Phase 3 — Keep as-is

Object	Action
Maintenance Work Order	Keep Migration Cockpit
Inventory EWM	Keep /SCWM/ISU or migrate to Migration Cockpit "Warehouse stock (EWM)"

6. Recommended Technical Pattern (BAPI + Package + Parallelism)

Reference implementation available at:

- Classic version (Clean Core Level 3): [Paralelismo/zbapi_open_po_parallel.abap](#)
- Modern version (Clean Core Level 1, ABAP Cloud + bgPF + RAP):
[Paralelismo/zcl_po_load_orchestrator.clas.abap](#)

Tuning parameters per profile

Profile	Package size	Parallelism	Scenario
Conservative	50	3	Production with online users
Balanced	100	5	Test / QA
Dedicated window	200–500	10–20	Weekend cutover

Applied principles

1. **Package of N records** per unit of work (no commit per record)
2. **BAPI_TRANSACTION_COMMIT WAIT = 'X'** at the end of the package (or **COMMIT ENTITIES** in RAP)
3. **STARTING NEW TASK DESTINATION IN GROUP** (aRFC) or **bgPF** (Level 1)
4. **WAIT UNTIL** or bgPF pool to limit concurrent tasks

5. **RECEIVE RESULTS callback** to collect the return from each package

6. **Synchronous fallback** on **RESOURCE_FAILURE**

7. Risks and Assumptions

Technical risks

Risk	Object	Mitigation
BAPI does not cover custom fields	Open Production Order, Open PO Conversion	POC before decision; complement with Z update
Concurrent locks under high parallelism	All	Adjust P_PARAL ; monitor SM12
Work process overload	All	Dedicated RZ12; monitor SM50
Mass SAPscript text (STXL, TEXT_SAP_*)	Prod Order, PO	Use native BAPI parameter instead of standalone SAVE_TEXT
Z tables outside BAPI scope	Open PO Conversion	Z update within the same package / same LUW
Pricing procedure configuration	Open PO Conversion	Confirm whether PRCD_ELEMENTS_PO is manual or determined

Assumptions

- Target system: SAP S/4HANA 2022 On-Premise (or compatible)
- Cutover window allows parallelism in dedicated mode
- Basis will provision RZ12 server group (**parallel_generators** or equivalent) or an appropriate bgPF profile
- Source data is already cleansed / reconciled (out of scope for this gain analysis)

8. Clean Core — Compliance Level

Approach	Clean Core Level	Cloud-ready?	Recommended for
LSMW / IDoc (current)	Level 3	No	Legacy, likely to be deprecated
BAPI + aRFC (Phase 1 proposal)	Level 3	No	Quick gain, on-premise system
BAPI encapsulated in class (Level 2)	Level 2	Partial	Bridge for future migration
RAP + bgPF + released APIs (Level 1)	Level 1	Yes	New cloud-first development

For the current scenario (reduce cutover window on S/4HANA 2022 On-Premise), **Level 3 with BAPI + parallelism is sufficient and pragmatic**. Migration to Level 1 can be done later without impacting the cutover.

9. Next Steps

1. Approve the phased plan
 2. POC of `BAPI_EQUI_CREATE` with a 10,000-record subset — measure actual time
 3. POC of `BAPI_PRODORD_CREATE` to validate coverage on Open Production Order
 4. Confirm objects available in the target system's Migration Cockpit (LTMC/LTMOM/Fiori app)
 5. Align with Basis: RZ12 server group, bgPF profile, work processes available during cutover
 6. Define log and reconciliation strategy per package (Application Log — `cl_bali_*`)
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